

2026 TURNAROUND MECHANICAL DECOKE

Cenovus Lima — Crude, Vacuum, HDS & Coker Furnaces

Job Report | USADebusk | Cenovus Lima Refinery — Lima, OH

PROJECT NO.	FACILITY	EXECUTION	PROJECT MANAGER
USA26046	Cenovus Lima Refinery	Sep 17 – 30, 2026	Jesse Utsey · Peter Campbell
WORK ORDERS	SCOPE	HEATERS	DURATION
Crude 82036534 · Vacuum 82036776 · Coker 82096135	TA Mechanical Decoke	4	14 days

4	501.5	378	3
HEATERS CLEANED	OPERATING HOURS	PIGS RUN	HEATERS SMART-PIGGED

Project Information

Customer Details

FACILITY	Cenovus Lima Refinery
ADDRESS	1150 S Metcalf St, Lima, OH 45804
WORK ORDERS	Crude 82036534 · Vacuum 82036776 · Coker 82096135
CONTACT	Alfredo Trevino (Crude, Vacuum, HDS) · Kaleb Dodson (Coker)

Project Details

SCOPE	2026 turnaround mechanical decoke of four fired heaters, with filtration and smart-pig support
EXECUTION	September 17 – 30, 2026 (14 days)
HEATERS	PR-170028 Crude · PR-170029 Vacuum Charge · PR-175169 HDS · PR-170002 Coker
EQUIPMENT	(2) Trimax Pumpers (Trimax 5 & 6) · (2) Support Units · (2) Filter Presses · Crew Trucks

Crew Details

PROJECT MANAGER	Jesse Utsey (Vacuum) · Peter Campbell (Crude, HDS, Coker)
SHIFT LEAD	Day — Jesse Utsey, Peter Campbell; Night — Sam Mixon, James McDaniel
DAYSHIFT	Danilo Ramirez, Blake Harman, Blake Baugh, Rodney Lynch, Brody Lebleau, David Charles, Adrian Garza, Marshall Douglas, Travis Trenholm
NIGHTSHIFT	Joel Ruiz, Tyler Garza, Justin Kelly, Mikee Green, Jeremiah Harman

Project Duration

SCOPE	UNIT	RIG	PIG	SMART PIG	SUBTOTAL
PR-170028 Crude Furnace	Trimax 5	35.75	118	38.25	192
PR-170029 Vacuum Charge Furnace	Trimax 6	58.5	63	74	195.5
PR-175169 HDS Furnace	Trimax 5	16	20.5	4.5	41
PR-170002 Coker Furnace	Trimax 5	22	51	0	73
TOTAL		132.25	252.5	116.75	501.5

Stand-By Summary

Stand-by is on-site time during which the pumping units were not actively pigging. Cause is recorded from the daily service receipts.

DATES	HOURS	CAUSE
Sep 19 – 21 (7077–7081)	49	Awaiting installation of the 12" radiant-outlet launchers (Vacuum)
Sep 26 (7091)	3	Awaiting relocation of the 12" launchers to passes 5, 7, 8 (Vacuum)
Sep 25 (8417)	1	Site safety review of hose routing near the catalyst job (HDS)
Sep 27 (8420)	1	Awaiting a 1600 CFM compressor to dewater (HDS)
TOTAL	54	

Combined project total: 501.5 operating hours plus 54 stand-by hours — 555.5 pumping-unit hours.

Pigs Used

SIZE	QTY	SIZE	QTY	SIZE	QTY
3"	3	3.8"	1	7.75"	3
3.125"	3	3.875"	4	8"	47
3.25"	6	4"	24	8.25"	56
3.3"	2	4.125"	15	8.5"	12
3.325"	3	4.25"	4	10"	20
3.5"	6	6"	44	12"	6
3.625"	21	6.125"	12		
3.75"	18	6.25"	68		
TOTAL	378				

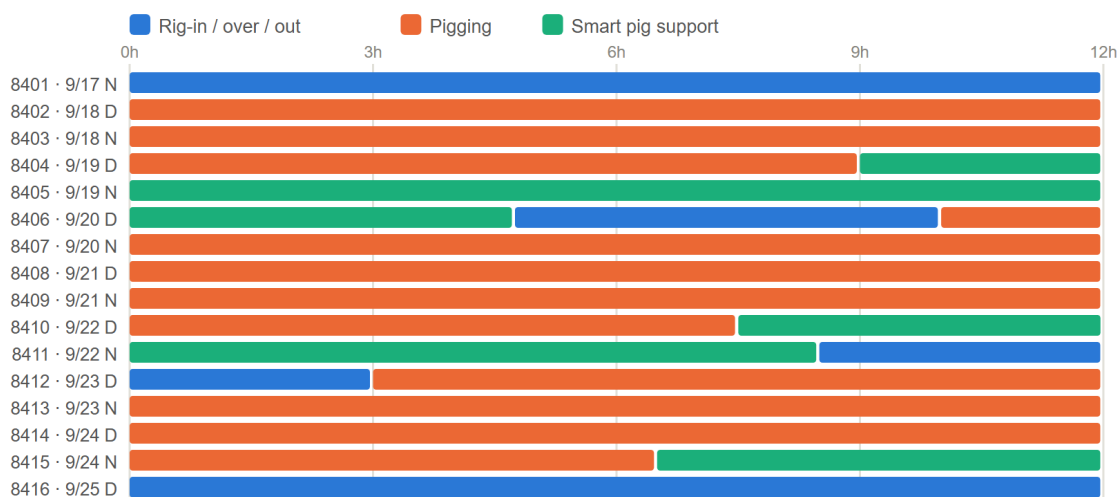
378 pigs across 22 sizes, 3" through 12".

Heater Data and Results

PR-170028 — Crude Furnace

Number of Passes	8, pigged individually
Total Footage	23,856 ft (2,982 ft per pass)
Convection Tube ID	5.761" (6" Sch 80S)
Radiant Tube ID	7.625" (8" Sch 80S)
Metallurgy	A312 TP317/317L
Return Bends	180° short-radius
Smart Pigging	Yes — Intero

Decoking Analysis: Most of the fouling was in the upper 8" radiant tubes, tubes 1 through 5. Average pigging time was about 39 hours per pass.

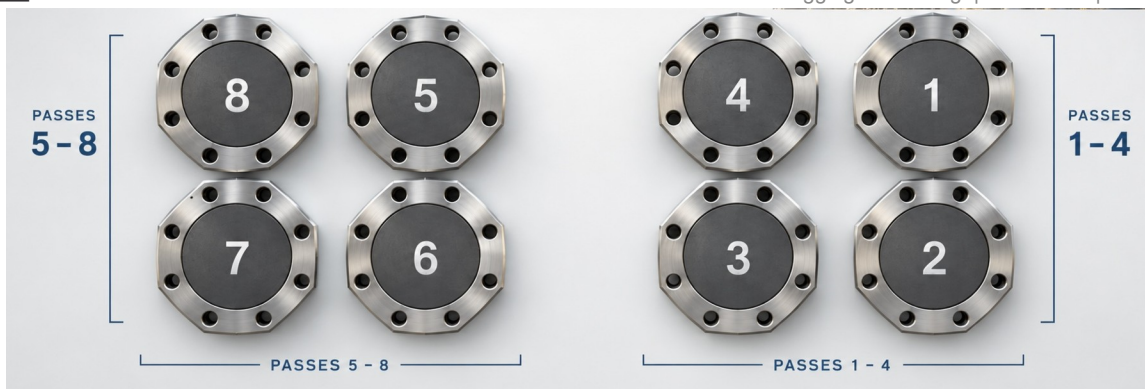


PR-170028 - task hours by shift, from the daily service receipts.

PR-170029 — Vacuum Charge Furnace

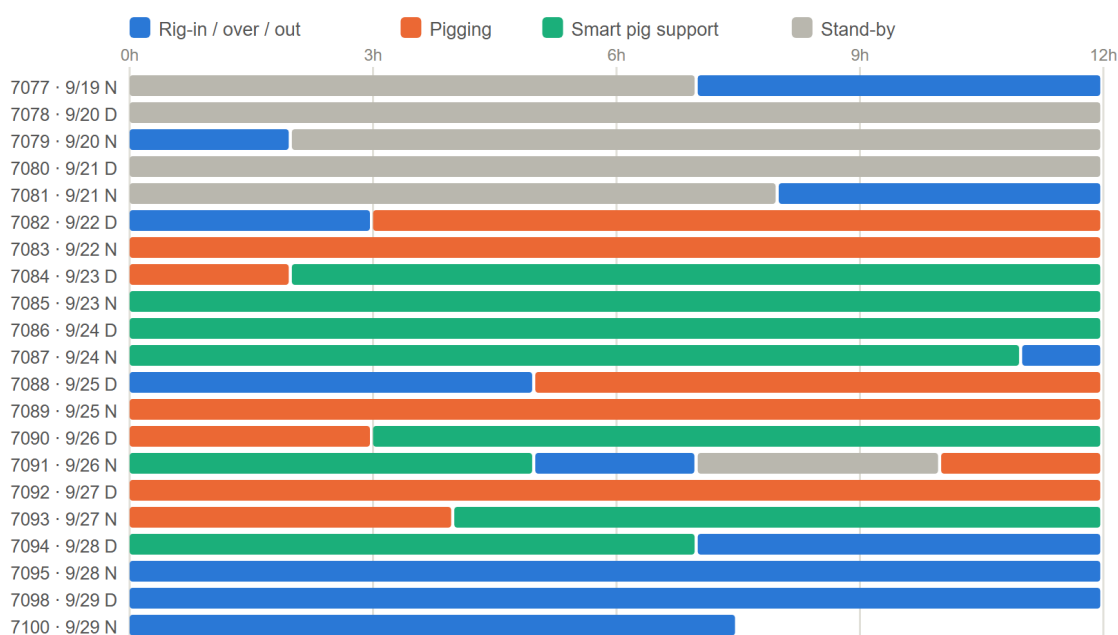
Number of Passes	8, pigged individually
Total Footage	7,054 ft (882 ft per pass)
Convection Tube ID	4.000"
Radiant Tube ID	3.826" · 5.761" · 7.813" · 9.750" · 11.750" (4" to 12")
Metallurgy	A312 TP317/317L
Return Bends	180° seamless
Smart Pigging	Yes — Intero

Decoking Analysis: Passes 4 and 5 were the dirtiest, pass 5 the most. Pass 5 had heavy fouling in the 12", moderate in the 10" and 4", and light in the 6" and 8". Pass 4 had moderate fouling in the 12", light to moderate in the 10", and light in the 8", 6" and 4". The other six passes were light throughout. The external 6" convection inlet piping, from the control valve to the 4" convection inlet tube, was heavily fouled on all eight passes, enough to stop 4" pigs entering the convection; it was cleaned first on every pass. Average pigging time was about 21 hours per pass.



Radiant outlet flanges, viewed facing the outlets. Each number is the pass, and the outlet belongs to the convection inlet of the same pass.

The chalk marks on the radiant outlet flanges and the flange tags were mislabelled. Use the layout above to match each radiant outlet to its convection inlet.

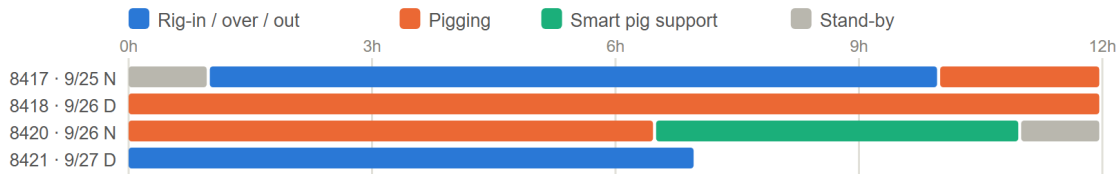


PR-170029 - task hours by shift, from the daily service receipts.

PR-175169 — HDS Furnace

Number of Passes	4, looped into 2 circuits (passes 1 + 2, passes 3 + 4)
Total Footage	3,600 ft (900 ft per pass)
Radiant Tube ID	5.885"
Metallurgy	A335 P5 (radiant)
Smart Pigging	Yes

Decoking Analysis: Light fouling throughout the convection and radiant sections. Average pigging time was about 20.5 hours per circuit.

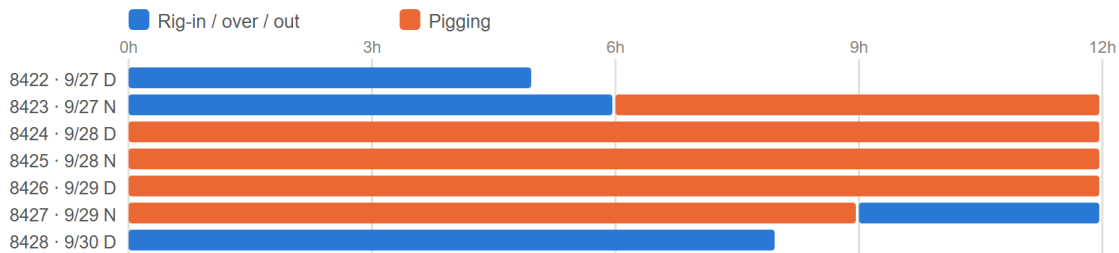


PR-175169 - task hours by shift, from the daily service receipts.

PR-170002 — Coker Furnace

Number of Passes	4, pigged as 3 circuits: convection (4 coils looped as one), radiant A + B, radiant C + D
Total Footage	15,904 ft (3,976 ft per pass)
Tube ID	3.398" (4" OD tube); radiant outlet section 3.624" (4" Sch 120)
Metallurgy	9Cr-1Mo
Return Bends	Cast 180° U-bends

Decoking Analysis: Light fouling in the convection, moderate in the radiant section. The crossovers were removed and the radiant passes looped A + B and C + D with temporary 180s at the radiant inlet flanges. Average pigging time was about 51 hours per circuit.



PR-170002 - task hours by shift, from the daily service receipts.

Flow Tests

PR-170028 Crude — Pass 1 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	310	270	40
700	285	255	30
650	250	225	25
600	210	180	30

PR-170028 Crude — Pass 2 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	315	280	35
700	295	270	25
650	270	255	15
600	230	210	20

PR-170028 Crude — Pass 3 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	350	305	45
650	275	220	55
550	220	165	55
450	160	120	40

PR-170028 Crude — Pass 4 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	324	295	29
650	248	230	18
550	177	165	12
450	123	115	8

PR-170028 Crude — Pass 5 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
750	270	245	25
650	215	190	25
550	155	125	30
450	100	85	15

PR-170028 Crude — Pass 6 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	300	285	15
700	280	255	25
600	240	205	35
500	215	185	30

PR-170028 Crude — Pass 7 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	290	265	25
700	265	240	25
600	230	205	25
500	200	175	25

PR-170028 Crude — Pass 8 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	250	221	29
700	220	190	30
600	200	165	35
500	160	100	60

PR-170029 Vacuum — Pass 1 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	295	220	75
650	260	185	75
600	225	165	60
550	200	135	65

PR-170029 Vacuum — Pass 2 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	275	225	50
650	235	195	40
600	205	170	35
550	160	130	30

PR-170029 Vacuum — Pass 3 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	332	282	50
650	290	250	40
600	255	220	35
550	220	190	30

PR-170029 Vacuum — Pass 4 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	280	210	70
650	235	188	47
600	210	160	50
550	170	135	35

PR-170029 Vacuum — Pass 5 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	270	220	50
650	235	190	45
600	202	160	42
550	170	140	30

PR-170029 Vacuum — Pass 6 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	285	225	60
650	235	195	40
600	210	165	45
550	180	140	40

PR-170029 Vacuum — Pass 7 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	260	220	40
650	225	185	40
600	195	160	35
550	160	120	40

PR-170029 Vacuum — Pass 8 (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	320	280	40
650	280	250	30
600	240	220	20
550	210	190	20

PR-175169 HDS — Passes 1 + 2 (looped) (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
800	320	267	53
750	290	238	52
700	250	215	35
650	215	184	31

PR-175169 HDS — Passes 3 + 4 (looped) (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
650	230	180	50
600	210	170	40
550	175	140	35
500	150	120	30

PR-170002 Coker — All convection (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
350	435	375	60
300	330	290	40
250	220	180	40
200	153	120	33

PR-170002 Coker — Passes C + D (Before | After)

GPM	BEFORE PSI	AFTER PSI	Δ PSI
350	419	287	132
300	320	219	101
250	225	159	66
200	142	108	34

Project Summary

USADebusk performed the mechanical decoke of four fired heaters for Cenovus at the Lima Refinery during the 2026 turnaround, September 17 – 30: the Crude Furnace (PR-170028), Vacuum Charge Furnace (PR-170029), HDS Furnace (PR-175169) and Coker Furnace (PR-170002). Two Trimax pumping units ran independent schedules — Trimax 6 on the Vacuum heater, and Trimax 5 on the Crude, HDS and Coker in turn. The Crude, Vacuum and HDS heaters were smart-pigged after cleaning.

Fifty-two of the project's 54 stand-by hours were on the Vacuum heater, waiting on the 12" radiant-outlet launchers to be installed and later relocated.

Project Close

All four heaters were cleaned and returned to Cenovus, and all equipment was demobilized from the facility on September 30, 2026.

USADebusk thanks the Cenovus Lima team for their coordination and support throughout the turnaround, and we look forward to continuing to serve your decoking and pigging needs.

Jesse Utsey

USADebusk

Project Manager

Images



Coke removed from the 12" tube, Vacuum pass 5.



The same pass 5 deposit, with a nut for scale.



Deposits removed from the external 6" convection inlet piping, collected from all eight Vacuum passes.